

Arduino-Based FM Receiver



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This project is of an FM radio based on Philips TEA5767 digital radio-receiver module. The radio receiver uses PC interface with Arduino UNO development board. The TEA5767 module offers such features as stereo or mono outputs, radio station scanning and signal strength indication. The author's prototype on a breadboard is shown in Fig. 1.

Circuit and working

The circuit of the Arduino-based FM receiver, shown in Fig. 2, is built around Arduino UNO board (board1), TEA5767 radio receiver module (FM1), LM386 low-power amplifier (IC1), 16x2 LCD (LCD1) and 8-ohm speaker (LS1).

Arduino UNO board. Arduino is an open source electronics prototyping platform based on flexible, easy-to-use hardware and software. It is intended for artists, designers, hobbyists and anyone interested in creating interactive objects or environments.

Arduino UNO is a board based on ATmega328 microcontroller. It has 14 digital input/output pins, six analogue inputs, a USB connection for programming the on-board microcontroller, power jack, an ICSP header and a reset button. It is operated with a 16MHz crystal oscillator and contains everything needed to support the microcontroller. It is easy to use as the user simply needs to connect it to a computer with a USB cable, or power it with an AC-to-DC adaptor or battery, to get started. The microcontroller on the board is programmed using Arduino programming language and Arduino development environment.

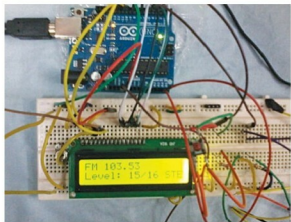


Fig. 1: Author's prototype

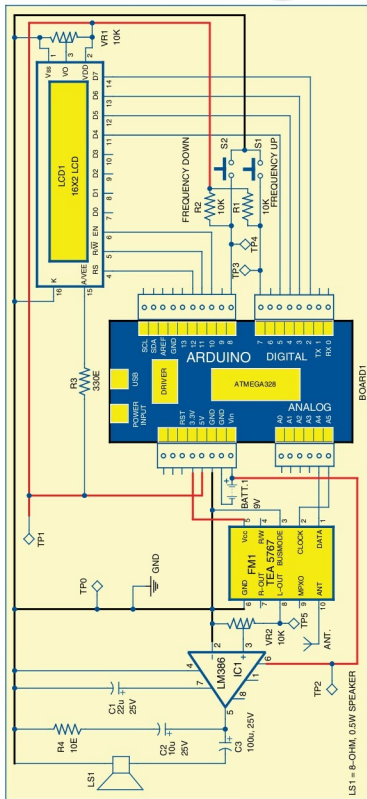


Fig. 2: Circuit diagram of the Arduino-based FM receiver